

Huu Thang Ly

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EDUCATION

University of Cincinnati (UC) | Cincinnati, OH Expected: May 2028
Master of Science in Computer Engineering
Florida International University (FIU) | Miami, FL May 2026
Bachelor of Science in Computer Engineering GPA: **3.97/4.00**

WORK EXPERIENCE

Global AI Mining Upland, CA (Remote)
Embedded AI Engineer Intern July 2025 - December 2025

- Developed a smart audio recording system using Raspberry Pi to enable real-time AI-powered monitoring, analysis, and response generation in live conversations, providing staff with instant answers to enhance service quality
- Researched deployment strategies for small language models on edge devices to reduce latency and support offline functionality for resource-constrained environments

Florida International University Miami, FL
Undergraduate Research Assistant / *Dr. Himanshu Upadhyay's Lab* September 2025 - December 2025

- Prototyped a real-time autonomous control pipeline integrating YOLO object detection and Gemini Live Model for dynamic environment response, validated on driving footage as a proof-of-concept for drone deployment.
- Architected multi-agent systems with LangChain for cybersecurity operations, delivering coordinated malicious link detection and security log analysis

Undergraduate Research Assistant / *Dr. Shekhar Bhansali's Lab* May 2024 - August 2025

- Built a VR-enabled 3D printing system with digital twin technology for remote printer monitoring and control, improving accessibility and efficiency in additive manufacturing
- Pursued research on advanced 3D printing with copper slurry and laser sintering to augment electrical conductivity and flexibility of sensors and printed circuit boards

PROJECTS

Flash-Recon | Senior Design | *Python, OpenCV, Open3D* August 2025 - April 2026

- Collaborated with 4 developers to build a real-time 3D reconstruction system, evaluated at 8–23 FPS across TartanAir, EuRoC, and TUM benchmarks using only standard monocular RGB cameras.
- Constructed and integrated a keyframe selection algorithm based on optical flow to filter redundant frames, reducing pipeline input volume by 40%.
- Tested and optimized bundle adjustment by creating a point cloud visualization to verify positional corrections and improve reconstruction accuracy.

InsuMAS | ShellHacks 2025 | *Python, LangChain, AI agents, Gradio* September 2025

- Partnered with 2 students to create a sophisticated multi-agent AI system providing comprehensive assistance for insurance-related queries
- Designed supervisor-agent architecture to intelligently route user requests, enabling the system to explain complex insurance terminology, recommend suitable plans, calculate costs, and locate healthcare providers

Butterfly Garden IoT | INIT Build | *C++, PlatformIO* October 2024 - December 2024

- Advanced FIU's first butterfly garden with a team of 10 contributors, utilizing IoT and embedded systems for environmental conservation and visitor engagement monitoring
- Engineered Arduino Uno and ESP32 microcontrollers to interface with temperature, humidity, air quality, and gas sensors for real-time environmental assessment

SKILLS

Programming: C/C++, Java, Python, JavaScript, HTML, CSS, MIPS, MATLAB, VHDL

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, OpenCV, Open3D, Langchain, React.js

Software: Git, Linux, Jupyter Notebook, Unity, Unreal Engine, Azure, Fusion 360, Power BI, Multisim, Vivado

Hardware: Analog/Digital Design, FPGA Development, Oscilloscope, Soldering, 3D Printing, Raspberry Pi, Arduino, ESP32

Certifications: Azure Fundamentals (AZ-900), AI Fundamentals (AI-900), CodePath Intermediate Web Development